

Natural History





About the Course

Learners dig deeper into relationships between creatures as they read about the beginnings of ecology. Coordinating afternoon activities are provided in Outdoor Work.

This topic is included in the following course(s): Science: Grade 7

About Science: Grade 7

In level 7, learners extend their relationship with science in a new direction (time), situating science in its historical, political, and cultural context, and beginning to understand that science is an ever-changing process rather than a static body of knowledge. Reading level and expectation increase from previous levels.

Science: Grade 7

The Big Picture:

To accomplish the goal of supporting a relationship with the Things of the Universe, a Mason science program consists of nature lore, natural history, and general science. Nature immersion, inquiry, community connection, and supportive literature are woven into each of these three parts. In Form 1, students became acquainted with this area of knowledge, science. Form 2 students extended their scope and began to encounter science as an active part of society. Form 3 students extend their relationship with science in a new direction (time) and consider science as a process by which man seeks Truth within Creation. They begin to situate science in its historical, political, and cultural context with all of the complexities of man's own story. In this way, students are prepared to think about science with a complete and holistic perspective.

Nature lore is timeless knowledge that is passed through a community, much like a grandmother passes on how to make that special bread when the dough just "feels right." Like Mason, we strive to pass on this knowledge primarily through outdoor work. Group nature walks, seasonal readings, and topics in scouting are provided as an Outdoor Work resource in the Quick Links. If desired, literature suggestions to support lore can be found in the Community Read Alouds resource (in Citizenship Grades 4+).



Placement & Combining Tips

About Science: Grade 7

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Natural History: Grade 7

For approximately seventh-grade students, based on a balance of complexity and reading level. However, learners may be freely combined based on needs and preferences.



Scheduling

GRADE	SCHEDULE INFO.	BOOKS
7	Natural History: Grade 7 1 time/week 30 min	The Girl Who Drew Butterflies: How Maria Merian's Art Changed Science

Sample Weekly View

Day 1	Day 2	Day 3	Day 4	Day 5
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Science: Grade 7

Natural History:
Grade 7

General Science:
Grade 7

General Science:
Grade 7

Nature Notebook:
Grade 7

Labs: Grade 7
Nature Walks &
Scouting: Grades 1-8



Planning & Prep

Permission to print for non-commercial use. See Alveary group use policy to use lessons in a group context.

LINKS: Click text or scan the QR code in the top corner of the lesson plan pages to view online resources associated with the lessons.

Responsibility for previewing all links rests with the teacher. All links were checked at the time of publication; however, websites change frequently and may contain objectionable content. Please report broken links by contacting us through our website.

Science: Grade 7

- ☐ Obtain any supplies indicated on the science or grade-level supply lists.
- ☐ Download any apps and shortcut any desired links.

Nature Lore:

- ☐ Bookmark your Outdoor Work Quick Link, so that you have it available on your weekly outing. Outdoor Work is generally flexible for your location and season and can be moved around in the schedule to incorporate or substitute Natural History Club outings, except where the suggestion is to collect some Thing for a lesson.
- ☐ Print or bookmark grade-specific nature notebook suggestions to support natural history and general science. Notebooking can be done on a walk, during occupations, or as a field trip, as appropriate.

Special Topics & Field Trips

Science: Grade 7

Encourage students to choose their own special topic and to notice its ecological relationships. What or who are they curious about or interested in getting to know better? Teachers can choose their own special study, too!

- ☐ Learn a few of your local species or varieties in connection with special topics.

Natural History: Grade 7

- ☐ Any term: natural history museum or butterfly house

Term Prep & Teacher Tips

Natural History: Grade 7

- ☐ Gather household items, typically easy for students to scavenge or teachers to obtain locally:
 - a flower with large, visible parts, such as a lily (Term 1)
 - a sunny window
 - water
 - a few sprigs of any water weed obtained from a pond or pet store that sells fish (Term 3)
 - various supplies by student design
 - optional: salt
 - optional: forceps/tweezers

Reminders

Natural History: Grade 7

- ☐ Note that the first lesson of Natural History is an activity, so be prepared with these and any other materials from the supply list on day 1.



Books & Resources

For book rationales and purchase options, click the Book List link or scan the QR code below.

∞ [View Book List Details](#)

Natural History: Grade 7



The Girl Who Drew Butterflies: How Maria Merian's Art Changed Science



Supplies

For supply list details and basic supplies helpful to have on hand, click the links or scan the QR code below.

∞ [View Basic Supplies](#)

∞ [View Supply List Details](#)

Natural History: Grade 7



Household Items - Natural History: Grade 7



Slides and Coverslips



Prepared Microscope Slides



Student Microscope



Quick Links

Related Course Quick Links

∞ [Extra Helpings](#)

∞ [Foundations \(See Section 13: Science\)](#)

∞ [Alternate One Year Physical Science Lesson Plans](#)

∞ [Outdoor Work](#)

∞ [Seek app from iNaturalist](#)

∞ [SkyView Lite for Android](#)

∞ [SkyView Lite for iOS](#)

Click THIS text
or scan the QR
code for links.



Natural History: Grade 7

How To Teach



Introduce

- Begin each lesson with learners' existing knowledge. If the book or activity is new or unfamiliar, then look at the title, a picture, or guidance in the lesson plan to help discuss what students think, drawing on previous experience. If continuing or revisiting a topic or activity, then recap.
- Often, the introduction in the lesson plans or the title of the book's section can help learners to draw out the main idea. Use this to support learners, as appropriate.
- Some learners may benefit from using pictures or looking back briefly.
- Allow them time to share any concerns and come alongside, as needed.



Read

- Read or do, as instructed in the lessons, noting any Teacher Tips provided. Learners should always have their journal, notebook, or other paper available in case they need to draw or diagram during the lesson.
- Use supportive strategies and educational tools to reduce frustration and better engage the mind, as appropriate. These could include, but are not limited to, the use of eBooks, pictures, audio, read-aloud, buddy reading, colored reading strips, etc.
- If learners do not understand a word or concept, do not worry. Try to show them with a picture or connect the idea to something they have seen in real life. They are learning much by the way and will likely build understanding over the term, the year, and beyond.



Narrate

- Process the ideas of the lesson by retelling events, describing, explaining a concept, etc. Tips about helping learners deal with various types of passages are provided, but teachers can learn more in Charlotte Mason's School Education Chapter 16, especially p.180.
- Learners may use words, pictures, Legos, PowerPoint, etc, to process and convey ideas in their own way.
- Teachers may take turns to model.



Discuss

- Consider together any thoughts, confusion, or concerns about the passage, keeping in mind that oftentimes new concepts are not going to be 'mastered' on the first or even second introduction. Mastery is not necessarily the goal, but curiosity and thinking.
- Questions/topics for further discussion are often provided in the lesson plans (or even lab books) to help. There are no right or wrong answers to these. Alternatively, many of these can be used for composition, depending on the needs of the learner and the instructional goals of the teacher. If teachers want to keep ideas active in the mind, these can also be used at other times to keep the ideas in the working memory.
- Note that learners may need to spend more than the allotted time engaging with or even repeating a lesson before moving on or as reinforcement at a later time. Adjust the pace as needed to feed the learner.
- Notice if there were any dates that they want to keep for their Book of Centuries.



Connect

- Follow any extra links, examine any sidebars in the text, look at pictures, etc., depending on learner needs and interest.
 - These can be viewed as alternative ways to engage.
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SAMPLE

Natural History: Grade 7

[Click THIS text or scan the QR code for links.](#)



Term 1

WEEK 1 30m Natural History: Grade 7 - Lesson 1

Shoots Activity

Materials: 9 bean seeds, 3 recycled pots or cups, soil of any kind, aluminum foil (for later), sunny window, recycled box to hold 3 pots or cups, water

→ INTRO

This term, you are going to read the biography of a woman who began as a young artist and grew to be a skilled naturalist and biologist. We will begin as she did, studying the anatomy and physiology of flowering plants. Today, you will plant some bean seeds and observe how they "see" the light.

→ ACTIVITY

- Prepare 3 pots or cups with soil and plant 3 seeds in each cup.
- Set the cups near a sunny window, but inside a recycled box on its side, such that they only receive light from the direction of the window. Keep them watered so that the soil is damp, but not wet. They may only need 1-2 tablespoons of water at a time, depending on the size of your cups. It will take them 1-2 weeks to germinate, and you will need to keep the soil damp.

→ NARRATE

Draw or diagram your planted seed cups in their window in your science notebook.

• OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 2 30m Natural History: Grade 7 - Lesson 2

The Girl in the Garden

Materials: The Girl Who Drew Butterflies

PREP: Read Teacher Tip

→ INTRO

Let's begin reading about Maria Merian today. As we continue to learn about her and her work, we will see the interactions between plants, insects, and the environment. These shared relationships exist for all living things around the world and are called the study of ecology.

→ READ, NARRATE, & DISCUSS

"Prologue" p.viii-ix "A girl kneels" - "her story."

★ TEACHER TIP

The next 2 weeks are activity weeks!

• OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 3 30m Natural History: Grade 7 - Lesson 3

Shoots Activity

Materials: aluminum foil, water

→ INTRO

You should have shoots by now, and they are probably growing rapidly! Today, you will change their conditions to see if they respond.

→ ACTIVITY

- Cut 3 small squares of aluminum foil (about 2" x 2") and 3 small rectangles (about 1/2" x 3").
- Form the squares over your fingertip or a pencil to form 3 small caps.
- Wrap the rectangles around your finger or pencil to form 1/2" tall tubes.
- Place the caps on the tips of the seedlings in one cup; place the tubes around the bases of the seedlings in another cup. Leave the third cup alone - this is your control. Continue watering your cups this week.

• OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

Natural History: Grade 7

[Click THIS text or scan the QR code for links.](#)



Term 1

→ NARRATE

Draw or diagram your seedlings in your science notebook.

SAMPLE